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Round-robin analysis of highly depleted lithium for Generation IV nuclear reactor applications

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ABSTRACT

Lithium reference materials containing unnaturally high abundances of ^{7}Li are not currently available, which poses quality control problems for highly depleted lithium materials (i.e., depleted in ^{6}Li) required for Generation IV nuclear reactors. This study presents an interlaboratory comparison of a lithium carbonate (NIST SRM924a) containing nominally natural isotopic abundances (~92.4 % Li-7) and a highly depleted lithium hydroxide material (~99.95 % Li-7). The natural lithium isotope abundances of NIST SRM924a are confirmed, and the ^{6}Li/^{7}Li ratio of the lithium hydroxide ranged from 0.000399 to 0.000436 with an average of 0.000428 ± 0.000023 (2SD, n = 9). Going forward this material can be used as quality control for analytical work involving highly depleted lithium.

1. Introduction

Several Generation IV nuclear reactor designs rely on the use of molten salt as a fuel carrier and/or coolant (Serp et al., 2014). Certain types of salt (e.g., FLiNaK, FLiBe) include lithium to reduce the viscosity of otherwise highly viscous molten salts, such as beryllium fluoride (Abe et al., 1981; Cantor et al., 1969). The drawback of the use of lithium in nuclear reactors is the production of tritium (Carpenter et al., 2017) that occurs by means of an (n,α) reaction, where ^{6}Li absorbs a neutron and undergoes alpha decay, producing a helium and tritium particle (Rumbaugh et al., 1938). Natural lithium contains isotopic abundances of 92.4 % ^{7}Li and 7.6 % ^{6}Li (Svec and Anderson, 1965); therefore, separation of the lithium isotopes to achieve ^{7}Li abundances appropriate for molten salt reactor operation, upwards of 99.995 % ^{7}Li, is desired.

Isotopic separation of lithium can be achieved through several different methods (see reviews by Symons, 1985; Murali et al., 2021; Badea et al., 2023). The COLEX process was originally used for separation of lithium isotopes that included mercury (Hg) amalgamation. This process has lost favorability due to energy use and environmental factors. New processes have been developed that are likely to supercede the COLEX process but have not been deployed at commercial scale. For example, liquid–liquid extractions using crown ethers have been tested for decades at the laboratory scale, which exploit the affinity of ^{6}Li over ^{7}Li to bind to different types of crown ethers in organic matrices (Pei et al., 2024; Fang et al., 2024; Zhang et al., 2022; Nishizawa et al., 1984). Electrochemical methods also exhibit feasibility due to differences in ion mobility between the two lithium isotopes (Wang et al., 2020; Ju et al., 2024; Bellora et al., 2023). Ion exchange resins also show promise due to differences in migration rates through resin columns that cause isotopic fractionations during Li elution (Suzuki et al., 2021; Hagiwara and Takakura, 1969). Regardless of the isotope separation method, the resulting products must be analyzed with high affinity to determine the effectiveness of the process.

Analysis of lithium isotope ratios in samples with nominally natural abundances is well established, while the analysis of isotopically pure lithium poses analytical challenges. These challenges are not significant for initial testing of isotope separation processes (i.e., when isotopic abundances are near natural), however, once isotopic compositions of

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materials exceed the range present in natural materials there are additional analytical considerations. Mass spectrometry analyses of lithium with artificially extreme isotopic ratios (here defined as abundance ratios > 1E3) are hindered by a lack of appropriate reference materials, as well as issues associated with laboratory contamination from the same element with natural isotopic abundances. The only currently available reference materials certified for isotopic abundances contain either nominally natural (LSVEC, IRMM-016) or ^{6}Li enriched (IRMM-615) isotopic compositions (Duta et al., 2009). Thus, a gap in quality control exists for ^{7}Li rich materials that are critical for modern nuclear energy applications.

This paper presents a round-robin measurement campaign to characterize two lithium materials, one containing nominally natural isotopic abundances (NIST924a) and another with a highly depleted (in ^{6}Li) isotopic composition. The highly depleted lithium material was purchased from IsoFlex USA, a supplier of various stable and radioactive isotopic materials. This material was the most depleted material commercially available (listed at 99.9607 % ^{7}Li), and no uncertainty is given on the certificate of analysis. The participating laboratories are Pacific Northwest National Laboratory (PNNL), Oak Ridge National Laboratory (ORNL), Idaho National Laboratory (INL), Lawrence Livermore National Laboratory (LLNL), and the University of Wisconsin-Madison State Laboratory of Hygiene (WSLH). The material provides a reference point for analytical efforts revolving around ^{7}Li enrichment for nuclear energy applications.

2. Methods

Primary solutions of each lithium material were prepared in a Class-1000 clean laboratory at WSLH. Aliquots of NIST924a (lithium carbonate) and the IsoFlex ^{7}LiOH (hereafter referred to as ^{7}Li Isoflex) were transferred into clean, pre-weighed 500 mL PFA bottles. Approximately 475 mL of 18.2 MΩ·cm H_2O were added to each bottle, followed by 25 total mL of concentrated (16 M) HNO_3. The HNO_3 was added in small increments due to the reactivity of the salts. The final solutions contained approximately 500 mL with 1000 and 2000 ug/g of Li in the ^{7}Li IsoFlex and NIST924a solutions, respectively. Five mL aliquots of each solution were transferred to pre-cleaned 5 mL centrifuge tubes for distribution to each laboratory. Additional aliquots were distributed to PNNL after the initial round for additional analyses. Isotopic ratio analyses were calibrated using standard reference materials available within each laboratory, including but not limited to IRMM-016, LSVEC, and/or internal reference materials with known isotopic compositions.

2.1. PNNL

2.1.1. Nu plasma 3

Samples were diluted to approximately 60 ppb total Li in 2 % nitric acid and analyzed on a Nu Plasma 3 multi-collector inductively coupled plasma mass spectrometer (MC-ICP-MS) equipped with an Apex Omega sample introduction system. The analysis was performed in static multi-collection mode, with ^{7}Li (signals between 5 – 40 V) placed on a faraday cup equipped with a 10^{11} Ohm resistor, and ^{6}Li (signals between 15 – 500 mV) placed on a faraday cup with a 10^{12} Ohm resistor. On-peak zeroes at atomic mass 6 and 7 were measured in the diluent reagent prior to the analysis of all lithium solutions for subtraction from the corresponding signals; these ranged between 0 and 30 mV due to the background interference from ^{14}N++ and ^{12}C++ ions (on ^{7}Li and ^{6}Li, respectively) and instrument Li background. Instrument mass bias was corrected for by measuring an in-house standard, CLWR-076 (^{6}Li/^{7}Li: 0.082121, used to support other DOE projects), and calculating a mass fractionation factor from the measured ratio (Eq (1)). Each sample analysis was bracketed by CLWR-076, and the measured ratio corrected by the average of the two factors calculated.

$$\left({}^{6}Li/{}^{7}Li\right)_{true} = \left(\left({}^{6}Li/{}^{7}Li\right)\right)_{measured} \times (6.015/7.016)^{FractionationFactor} \quad (1)$$

2.1.2. Neoma MS/MS

Primary solutions were each diluted using 3 % (w/w) by a factor of 1000 to provide 1 and 2 ppm solutions of the ^{7}Li IsoFlex and NIST924a, respectively. These solutions were then further diluted to provide 6 ppb each for isotopic analysis. A ThermoScientific Neoma MS/MS (Bremen, Germany) MC-ICP-MS equipped with 11 faraday cups, a center SEM with RPQ, and a low-mass nuclear ion counting package, was used for isotopic determination. Li isotopes were collected using L5 (^{6}Li) and H5 (^{7}Li) cups, with a center cup mass of 6.7 AMU. The double Wien filter was tuned to reduce spread of the ion beam with the magnetic field set to 5 % and the electric field voltage set to ~ 150 V. The L5 cup was equipped with a 10^{13} Ohm resistor, and the H5 cup was equipped with a 10^{11} Ohm resistor. Samples solutions were introduced into the plasma using a 100 uL/minute PFA nebulizer attached to an ApexOmega desolvating nebulizer. Isotopic data was collected using 1 block of 30 cycles. Each cycle was integrated for 4 sec. On-peak baselines were measured in the same way before and after every sample and standard analysis. Peak centering occurred throughout analytical sequences. Mass bias was corrected by sample-standard bracketing using isotopically certified material LSVEC with a ^{6}Li/^{7}Li = 0.08215 ± 0.00023 (Qi et al., 1997).

2.1.3. iCAP TQ

Primary solutions were each diluted using 3 % (w/w) by a factor of 1000 to provide 1 and 2 ppm solutions of the ^{7}Li IsoFlex and NIST924a, respectively. These solutions were then further diluted to provide 1 ppb each for isotopic analysis. A ThermoScientific iCAP TQ (Bremen, Germany) inductively coupled plasma mass spectrometer (ICP-MS) equipped with a single secondary electron multiplier was used for isotopic determination. Lithium isotopes were measured in single quadrupole mode with beam intensities below the pulse counting mode threshold. Samples solutions were introduced into the plasma through a peristaltic pump attached to a PFA nebulizer and single pass cyclonic spray chamber housed within a Peltier cooler. Isotopic data were collected using 10 replicates of 50 sweeps with integration times of 0.01 and 0.12 s on the ^{7}Li and ^{6}Li beams, respectively. On-peak baselines were measured in the same way before and after every sample and standard analysis. Mass bias was corrected by sample-standard bracketing using isotopically certified material LSVEC with a ^{6}Li/^{7}Li = 0.08215 ± 0.00023 (Qi et al., 1997).

2.1.4. Triton

The PNNL Thermal Ionization Mass Spectrometry (TIMS) method is based on the procedure outlined in (Moriguti and Nakamura, 1993). 25 ng of lithium carbonate standard was dried down onto degassed Re double-filaments. 5 % phosphoric acid was deposited onto the filament to create a lithium phosphate ion source. The current on the filament was increased under atmosphere to 1.9 A until excess phosphoric acid fumed from the filament and a slight under-glow was observed.

The lithium carbonate standards were analyzed on a Thermo Scientific Triton Plus using Faraday Cups H4 and L4 and 10^{11} Ohm resistors for ^{7}Li and ^{6}Li, respectively. The samples were analyzed at a source vacuum pressure of < 1E-7 mbar. The sample filaments were ramped at a rate of 15 mA/min to a current of approximately 800 mA; the ionizer filament was maintained at approximately 1700 mA. All standards were analyzed with the ^{7}Li isotope at an intensity of > 30 V (300 pA). A large ion beam was necessary to measure > 10 mV of ^{6}Li on the isotopically depleted ^{7}Li Isoflex standard. Sixty ^{6}Li/^{7}Li ratios were measured per analysis. All data was baseline and gain corrected using the commercial Thermo Scientific Triton Plus software.

Lithium carbonate standard LSVEC (certificate value ^{6}Li/^{7}Li = 0.08215 +/- 0.00023 2σ) was used to calculate an external mass bias correction factor. Six replicates of LSVEC yielded an uncorrected ^{6}Li/^{7}Li value of 0.08320 +/- 0.00047 2σ. The calculated mass bias correction factor of 1.0021 +/- 0.0057 2σ was applied to all other lithium

carbonate standards. The standard deviation of the six LSVEC analyses (~0.57 % 2σ) was propagated into the expanded uncertainty of the IRMM-016, ^{7}Li Isoflex and NIST 942a lithium carbonate ^{6}Li/^{7}Li values.

2.1.5. Nuclear magnetic resonance

Nuclear Magnetic Resonance (NMR) analysis was tested to determine feasibility and sensitivity for analysis of depleted lithium materials. Aliquots of NIST924a and ^{7}Li IsoFlex solid materials (not the same solutions as distributed for mass spectrometry analyses) were prepared for the NMR analyses. To determine the isotope ratio in the ^{7}Li IsoFlex, a series of solutions were prepared in which the NIST924a Li_2CO_3 was added to ^{7}Li IsoFlex solutions in nitric acid modified by Fe^{3+}. Iron nitrate nonahydrate (4 mM) was dissolved in 3.6 wt% nitric acid. The ^{7}Li IsoFlex was then added; note that given potential for the mass to be affected by the hydration state of $LiOH{\cdot}H_2O$ and the extent of surface-bound water and/or anhydrous LiOH, the mass was not used to determine the concentration in the isotope ratio determination and instead the method of standard addition was used to determine the concentration of the ^{7}Li IsoFlex only sample spectroscopically. Then, the ^{7}Li IsoFlex solution in nitric acid modified by iron was used to prepare an array of samples in which known amounts of NIST924a were added (standard addition method). The isotope ratio was assumed to be that of the average of all mass spectrometry analyses of NIST924a from this study. These solutions were characterized via single pulse direct excitation ^{6}Li and ^{7}Li NMR spectroscopy. Additional details are provided in the supporting information.

2.2. ORNL

Optima™ HNO_3 and HF were purchased through Fisher Scientific and used without further purification. ASTM Type I (18.2 M OHM) water was generated with a Thermo Scientific™ Barnstead™ GenPure™ Pro Water Purification System. Single element standard of Li was purchased from High Purity Standards and used without further purification. Certified Reference Materials (CRMs) for Li isotopic compositions were purchased from the Institute for Reference Materials and Measurements [IRMM-016], now the Joint Research Center of the European Commission (JRC-EC), or National Institute of Standards and Technology (NIST) Standard Reference Material for SRM 3129A (Code 312) and 8545 (Code 845). An internal Oak Ridge National Laboratory (ORNL) solution of ^{6}Li (15 %, Code 151) was used as an additional isotopic QC standard.

To prepare the non-digested sample, a 4 μL neat fraction was withdrawn and diluted to 2 mL with 2 % (w/w) optima grade nitric acid. An additional dilution was made to yield a signal of 16 V on the H4 (^{7}Li) faraday cup. The second dilution was approximately 50 μL diluted with 2 % (w/w) optima grade nitric acid to a final volume of 2 mL. The exception was Sample 6 which was diluted to a target intensity of 45 V on the H4 (^{7}Li) faraday cup. The digested samples were prepared by removing a 100 μL fraction and with 5 mL of 16 M (w/w) Optima grade nitric acid into a 35 mL pyrex microwave vessel with a PFA liner. The samples were digested using a CEM Discover microwave with stirring at 200°C for 10 min. The microwave vessels were cooled with purified air and quantitatively transferred to a 15 mL centrifuge tube for analysis.

Both the non-digested and digested samples were analyzed on three separate days. Each day of analysis used a unique instrument calibration, zoom lens settings, and tuning. A ThermoScientific Neoma (Bremen, Germany) MC-ICP-MS equipped with 9 faraday cups and center SEM with RPQ was used for isotopic determination. Li isotopes were collected used L4 and H4 cups, with a center cup mass of 6.465 AMU. Samples solutions were introduced using a 100 uL/minute integrated PFA nebulizer coupled to a scott double pass spray chamber. Isotopic data was collected using 1 block of 30 cycles. Each cycle was integrated for 16 seconds. The first line was used to peak center before data collection. Samples were bracketed between mass standards and QC Standards.

2.3. INL

Primary sample solutions were diluted to approximately 100 ppb. Lithium isotopes (^{6}Li and ^{7}Li) were both measured on faraday cups (L4 and H4) on a ThermoScientific Neptune MC-ICP-MS. Samples were introduced into the plasma using an ElementalScientific Apex desolvating nebulizer. Samples were bracketed with blanks and LSVEC in sequences that comprised Blank-LSVEC-Blank-Sample-Blank-LSVEC. Samples were background subtracted by bracketing blanks and corrected for mass bias by bracketing LSVEC standards. Each measurement consisted of 10 cycles of 4 s integrations, and the sample sequence was run in triplicate.

2.4. LLNL

Lithium isotopic analyses were performed using the Thermo Scientific NeptunePlus MC-ICP-MS at LLNL. The instrument was fitted with the high sensitivity 'Jet' sample cone and the regular 'H' skimmer to boost sensitivity and optimize peak shape. The two isotopes of lithium are measured on the L4 and H4 Faraday collectors assigned with 10^{11} Ω resistors and using a dummy mass on the center cup. A total of four separate analytical sessions were performed on separate days to characterize the isotopic composition of the two reference materials. Sessions 1, 2 and 3 used an ESI Apex 2IR desolvating nebulizer to introduce sample solutions to the plasma, whereas Session 4 used a Cetac Aridus. Samples and standards were diluted to similar running concentrations (± 10 %) with final Li concentrations of 50 ng/g (Session 1 and 2), 150 ng/g (Session 3) and 100 ng/g (Session 4). As there are few polyatomic or isobaric interferences to consider at the lithium mass range the isotopic analyses were performed at low resolution. Typical sensitivity using the Apex 2IR and an ESI 50 μl/min Teflon self-aspirating nebulizer was ~ 140 V/ppm. A higher sensitivity of ~ 400 V/ppm was achieved using the Aridus and a Cetac 100 μl/min nebulizer. Each individual isotopic analysis consists of 30 × 8 s integrations, with raw voltages being corrected for instrumental background using the preceding blank analysis. The ^{7}Li background was usually 20–30 mV, equating to signal to background ratios of 300–1000 in all samples and standards. The ^{6}Li background was ~ 2 mV, which is significant (20–50 %) for the ^{7}Li Isoflex sample, in which ^{6}Li comprises ~ 0.04 % of the sample. Despite this, replicate isotopic analyses of the ^{7}Li Isoflex sample produced relatively consistent ^{7}Li/^{6}Li ratios. Typical internal precision on the ^{7}Li/^{6}Li ratio ranged from 0.01 to 0.05 ‰ for natural-like samples/standards and 0.6–1.5 ‰ for ^{7}Li Isoflex.

2.5. WSLH

A ThermoScientific NeptunePlus MC-ICP-MS equipped with 9 faraday cups, a center SEM with RPQ, and three low mass compact discrete dynodes, was used for isotopic determination. Two collector configurations were tested. The first measured Li isotopes using L4 and H4 Faraday cups, both equipped with 10^{11} Ohm resistors, and a center cup mass of 6.57 AMU. The second method measured ^{7}Li on the H4 faraday cup, and ^{6}Li on the IC5 as described in (Scott and Shafer, 2020). Samples solutions were introduced using a 100 uL/minute integrated PFA nebulizer coupled to a scott double pass spray chamber. Isotopic data was collected using 1 block and 10 cycles. Each cycle was integrated for 4 seconds. Mass bias was corrected using sample-standard bracketing with LSVEC as the primary standard. On-peak baselines were measured before and subtracted from every sample and standard analysis.

3. Results

The ^{6}Li/^{7}Li of the ^{7}Li Isoflex material ranges from 0.000399 to 0.000436 with an average of 0.000428 ± 0.000023 (Table 1 and Fig. 1). The ^{6}Li/^{7}Li of the NIST924a lithium carbonate ranges from 0.082098 to

Table 1
Lithium isotope analysis results for NIST SRM924a.

LAB	Instrument	$^{6}Li/^{7}Li$	Uncertainty (2σ)	^{7}Li Atom %	Uncertainty (2σ)	Replicates
ORNL	Thermo Neoma	0.082278	0.000095	92.398	0.008	18
LLNL	Thermo Neptune	0.082270	0.000230	92.398	0.019	9
PNNL	Nu Plasma3	0.082098	0.000018	92.413	0.002	6
PNNL	Thermo Neoma MS/MS	0.082282	0.000231	92.397	0.020	3
PNNL	Thermo iCAP	0.082177	0.000413	92.406	0.035	6
PNNL	Thermo Triton	0.082271	0.000560	92.398	0.048	6
INL	Thermo Neptune	0.082280	0.000022	92.398	0.002	3
WSLH	Thermo NeptunePlus (F)	0.082272	0.000233	92.398	0.020	9
WSLH	Thermo NeptunePlus (F-IC)	0.082168	0.000304	92.407	0.026	6
Average		0.08223	0.00013	92.402	0.012	

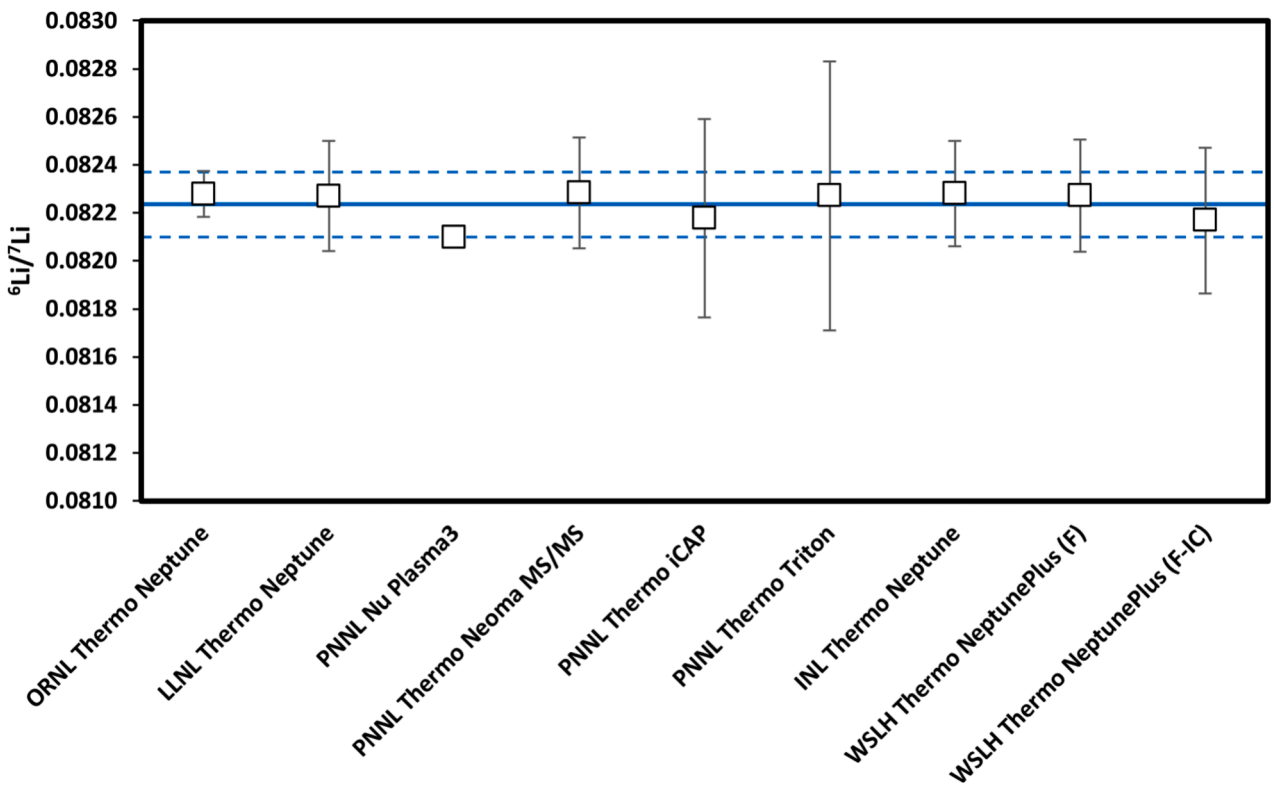

Fig. 1. Graphic presentation of lithium isotope results for NIST SRM924a.

0.08228 with an average of 0.08223 ± 0.00013 (Table 2 and Fig. 2). The quoted uncertainties for both materials represent 2 standard deviations of the mean value calculated from 9 separately reported values and does not account for any interlaboratory biases. The NMR analyses found ^{6}Li below detection and gave a $^{6}Li/^{7}Li$ value of < 0.001. Additional details of the NMR results are provided in the supplementary file. Uncertainties reported from each laboratory include all relevant sources of error and are mostly constrained by the uncertainty in the isotopic composition of the primary standard used for sample-standard bracketing, or in the case of the TIMS the fractionation factor determined from multiple measurements of LSVEC. All analytical results produced % ^{7}Li abundances of the ^{7}Li IsoFlex material below the abundance listed on the certificate of

Table 2
Lithium isotope analysis results for 7Li IsoFlex.

LAB	Instrument	$^{6}Li/^{7}Li$	Uncertainty (2σ)	^{7}Li Atom %	Uncertainty (2σ)	Replicates
ORNL	Thermo Neoma	0.000436	0.000092	99.9564	0.0088	12
LLNL	Thermo Neptune	0.000399	0.000022	99.9602	0.0023	9
PNNL	Nu Plasma3	0.000424	0.000003	99.9576	0.0004	7
PNNL	Thermo Neoma MS/MS	0.000432	0.000004	99.9568	0.0004	3
PNNL	Thermo iCAP	0.000435	0.000023	99.9565	0.0023	6
PNNL	Thermo Triton	0.000429	0.000005	99.9571	0.0004	5
INL	Thermo Neptune	0.000429	0.000005	99.9571	0.0005	3
WSLH	Thermo NeptunePlus (F)	0.000433	0.000009	99.9567	0.0009	41
WSLH	Thermo NeptunePlus (F-IC)	0.000432	0.000004	99.9568	0.0004	6
Average		0.000428	0.000023	99.9573	0.0023	

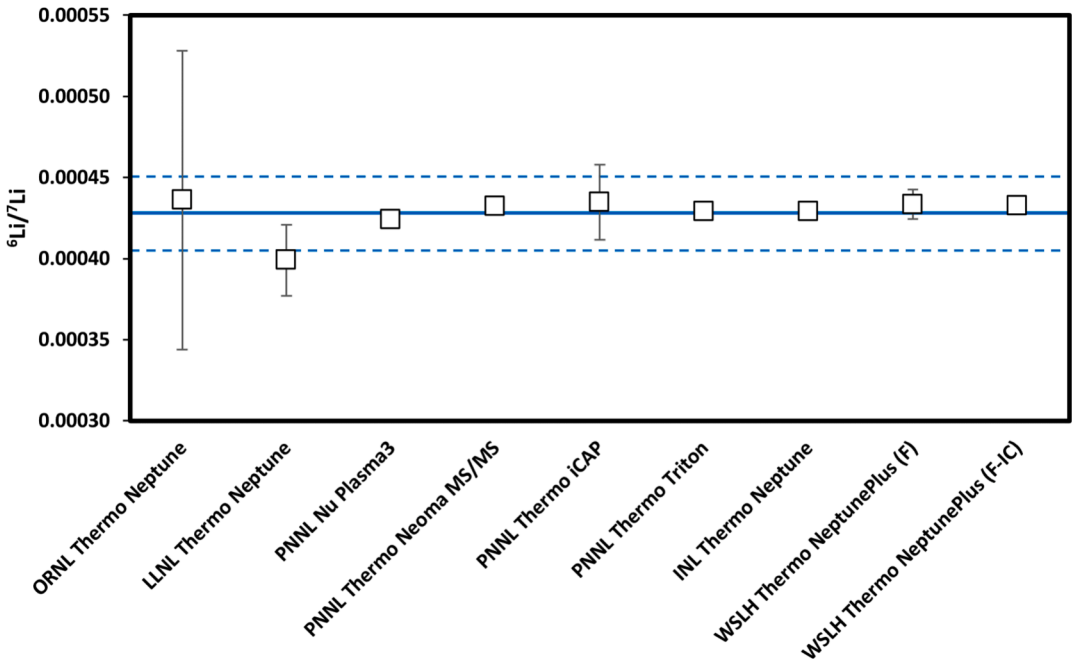

Fig. 2. Graphic presentation of lithium isotope results for ^{7}Li IsoFlex.

analysis. The NIST924a material contains nominally natural isotopic abundances as quoted in the certificate of analysis.

4. Discussion

4.1. Mass spectrometry analyses of lithium isotopes

The accuracy and precision of the lithium isotopic abundance measurements is limited by the uncertainties in the certified isotopic compositions of the reference materials used for mass bias corrections. Given those uncertainties, all isotopic results are generally within uncertainty of one another. The reproducibility of the NIST924a standard represented by 2 standard deviations of the average value for all labs is comparable to uncertainties of the primary standards LSVEC and IRMM-016 that were used for the mass bias correction.

One of the outstanding questions that was not addressed in this study is the potential impacts of using a reference standard with significantly different isotopic composition than the "sample" to correct for mass bias. Good agreement between mass spectrometry methods (TIMS and ICP-MS) suggest that whatever potential fractionation bias exists is consistent. No isotopically certified standard reference materials with ^{7}Li abundances significantly greater than that found in nature are commercially available. An intention of this work is to provide a consensus value for a commercially available material that could be used for future studies that include measurement of ^{7}Li rich material. Use of such a material could benefit analyses of additional lithium materials with greater purity targets by reducing the effects of natural background lithium on the ^{6}Li measurement.

In addition to instrumental backgrounds from natural lithium that have an outsized impact on the ^{6}Li abundance, other instrumental backgrounds can also affect isotopic composition measurements. For example, on the Nu Plasma 3 the backgrounds from ^{14}N++ and ^{12}C++ can interfere on the ^{7}Li and ^{6}Li ion beams of the sample. This is generally not an issue for natural lithium samples with relatively large ion beams for both lithium isotopes but can affect isotopic compositions for samples with unnatural lithium isotope abundances. These backgrounds should be measured prior to the sample analysis and subtracted from the sample beam intensities as was done in this study. Other ICP instruments may suffer from the same issues. Both the Neptune and Neoma instruments can resolve the ^{14}N++ background on ^{7}Li, and a ^{12}C++ background was not observed near the ^{6}Li mass position (Fig. 3). On a TIMS instruments, such as the Triton, these backgrounds are a non-issue due to these elements being volatilized off the filament prior to ionization and analysis of Li. In addition, unlike the ICP source at atmospheric pressure, the source of the TIMS is under vacuum and these elements are not introduced into the mass spectrometer. Nevertheless, the background count rates on the detectors used for lithium isotopes should always be characterized regardless of the type of mass spectrometer being used.

4.2. Implications for operation of Generation IV nuclear reactors

Results of this work show that depleted lithium can be well characterized using several different types of mass spectrometers. Mass spectrometric analyses of materials being using for nuclear reactors should be required to validate the isotopic compositions of those materials and ensure that information provided by the source is accurate. This could become especially important for situations where the isotopic composition is more extreme than that measured in this study, such as for molten salt reactors that typically use an order of magnitude greater ^{7}Li abundance in the source material than what was measured in the ^{7}Li IsoFlex (Rosenthal et al., 1971). If the isotopic composition of the Li-bearing reactor materials does not meet specifications, then the production of tritium during operation would be higher than expected. The production of tritium is an environmental concern; therefore, confirmation of the Li isotopic composition of reactor materials is warranted.

Validation of the isotopic compositions of reactor materials would benefit from the availability of certified reference materials with similar isotopic compositions to those expected in such materials. The material used for this round robin study was not exhausted during the study and can therefore serve as an interim reference material. However, efforts to produce an isotopically certified reference material with depleted ^{6}Li/^{7}Li in the range required for molten salt reactors should be pursued. Certification of such a material should include multiple mass spectrometry analyses from independent laboratories to provide a robust isotopic composition and associated uncertainty.

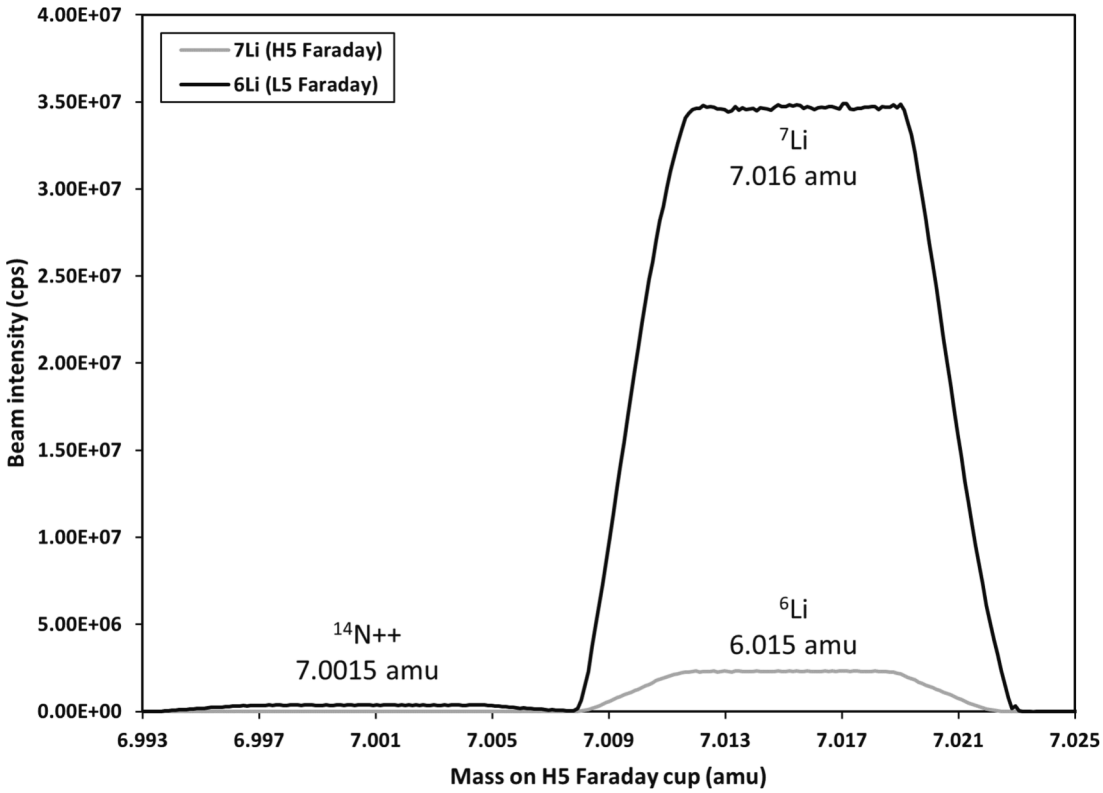

Fig. 3. Mass scan of a 500 ppt LSVEC solution on the Neoma MS/MS MC-ICP-MS. The double charged nitrogen peak (14 N++) is fully resolved from the ^{7}Li peak on the H5 Faraday cup.able.

5. Conclusions

Here we presented lithium isotopic abundance data for two materials, NIST924a and a highly depleted LiOH purchased from IsoFlex (^{7}Li IsoFlex). Results from 9 different methods on 8 different mass spectrometers gave ^{6}Li/^{7}Li values of 0.08223 ± 0.00013 in NIST924a and 0.000428 ± 0.000023 in ^{7}Li IsoFlex. The nominally natural lithium isotope abundances of NIST924a are confirmed. The ^{7}Li IsoFlex material can be used as a reference value for future studies requiring analyses of highly depleted Li materials for nuclear energy applications. The instruments, methods, and materials presented in this study could be used to validate the isotopic composition of materials used in nuclear reactors.

CRediT authorship contribution statement

Sean R. Scott: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Johnny Williams:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Sara Mastromarino:** Writing – review & editing, Methodology, Conceptualization. **Norbert Gajos:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Christian Berry:** Methodology, Formal analysis. **Ian Anderson:** Methodology, Formal analysis. **Steven Shen:** Methodology, Formal analysis, Data curation. **Trent R. Graham:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Cole Hexel:** Methodology, Formal analysis. **Josh Wimpenny:** Writing – review & editing, Methodology, Formal analysis. **Jacob Brookhart:** Methodology, Formal analysis. **Alan Kruizenga:** Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nucengdes.2024.113664.

Data availability

The data is included in the paper

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